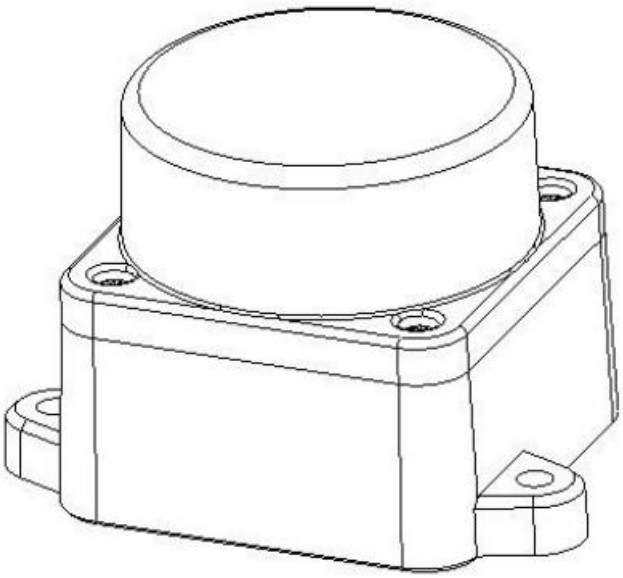
国科光芯
CHINA SCIENCE PHOTON CHIP

COIN-D6

Development Manual



V1.0

Catalogue

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Data format standard specification

1、Radar operating mode

- The COIN-D6 LiDAR supports command-controlled start and stop functions. After power-on, the radar is in a state of stopping scanning and ranging, and can be started and stopped through commands at this time

  Start：

  Send:0xAA 0x55 0xF0 0x0F

  Ok Ack:0xA5,0x5A,0x50,0x07,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0xA8

  Error Ack: 0xA5,0x5A,0x55,0x07,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0xE9

  Stop：

  Send: 0xAA 0x55 0xF5 0x0A

  Ok Ack: 0xA5,0x5A,0x55,0x07,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0Xad

  Error Ack: 0xA5,0x5A,0x55,0x07,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0xE9

- The COIN-D6 LiDAR only outputs data after its rotational speed stabilizes. During the speed adjustment process, it emits double-byte speed adjustment commands accompanied by some 0xFE or 0xFF。
- The COIN-D6 LiDAR undergoes closed-loop adjustment through an internal speed regulation module, maintaining a fixed rotation speed of 10Hz.

2、Radar command format



A5 5A ：Frame Header

XX XX ：Length, little-endian, e.g., 05 00 represents length 5.

YY YY ：Checksum, the sum of all other bytes

ZZ ：Packet type: 0x81 = Upload LiDAR data; 0x01 = Upload LiDAR information.

：Data field

2.1、LiDAR device information command

LiDAR information upload. This command is the first command sent by the LiDAR after power-on, and it provides a detailed description of the information of the COIN-D6 LiDAR device.

A5	5A	14	00	E3	02	01	Data field

The data field length is 20 bytes, and the specific data is as follows.:

Byte offset:

1		3		5		7		9		11		13		15		17		19	
43	4F	49	4E	2D	44	34	00	00	00	00	00	00	00	00	00	00	00	00	01

Bytes 1-12 are the LiDAR model. Among them, bytes 1-7 are the ASCII code for COIN-D6; bytes 8-12 are reserved.

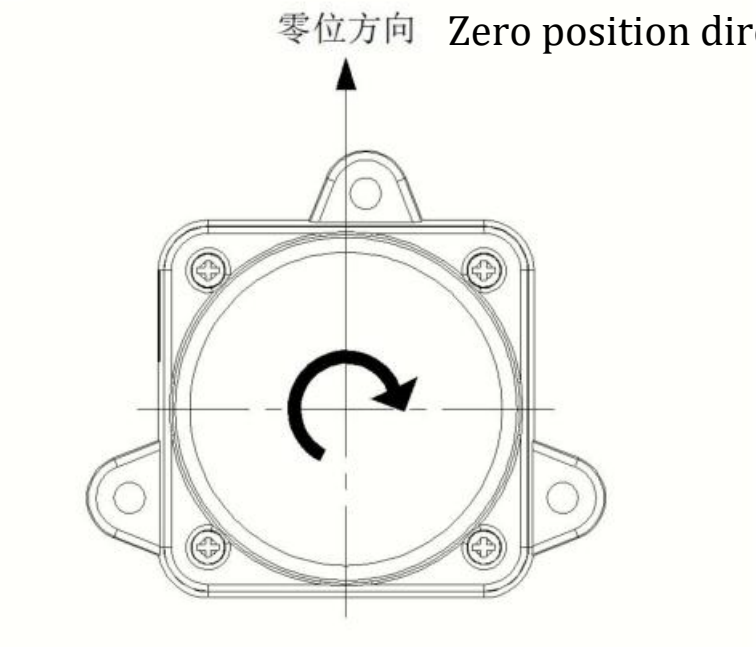
Bytes 13-14 are 00 00, representing the deviation between the LiDAR data zero angle and the identification zero angle. The COIN-D6 has already completed angle compensation structurally, so ignore this compensation value.

Byte 15 is 0x00, indicating that the LiDAR rotation direction is clockwise.

Byte 16 is 0x01, indicating that angle correction needs to be implemented in the robot's driver code.

Bytes 17-19 are reserved and default to 0.

Byte 20 is 0x01, indicating that the current software version number is Rev 1.



2.2、LiDAR data command

Start scan. This command is sent fixedly before starting the scan,

and it is the second command sent by the LiDAR after powering on.

A5	5A	00	00	80	01	81	Data field

Where 0x80 0x01 is the checksum, and 0x81 is the type code.

The data field is the point cloud data scanned by the system, and the unified data format is as follows.

PHL	PHH	M&T	LSN	FSAL	FSAH	LSAL	LSAH	CSL	CSH	S1_L	S1_2nd	S1_H	S2_L	S2_2nd	S2_H	……
AA	55															

Point cloud data format parsing

Identifier	Definition	Description
PH(2B)	Header	Fixed as 0x55AA, little-endian mode.
M&T(1B)	Scan frequency and data packet type	M-bit1:bit7: indicates the current scan frequency of the radar, valid in the start data packet, defaults to 0 in point cloud data packets; T-bit0: indicates the type of the current data packet; 0x00: point cloud data packet, 0x01: start data packet.
LSN(1B)	Number of sampling points	Indicates the number of sampling points contained in the current data packet; there is only 1 start point data in the start data packet, and this value is 1.
FSA(2B)	Start angle	Angle data of the first sample, little-endian.
LSA(2B)	End angle	Angle data of the end sample, little-endian.
CS(2B)	Checksum	Checksum of the current data packet, verified by performing a two-byte XOR on the current data packet.
Si(3B)	Sampling data	System test sampling data, consisting of distance and intensity values of sampling points.

M&T Parsing:

M: Indicates the current scanning frequency, M = M & [binary] >> 1;

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T: Indicates the data packet type; T = (M & T) & 0x01; When T=1 is detected,

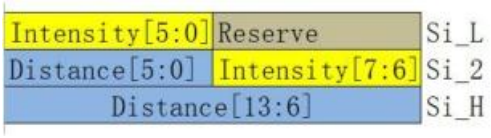
it indicates that this packet is the start data packet, representing the

beginning of a full scan of point cloud data. In this data packet, LSN = 1,

which means the quantity of Si is 1;

Si Parsing:

Si information includes distance values, intensity values, and other information of the sampling points. The specific format definition is shown in the figure below. The specific action is defined as:



Intensity Parsing:

Intensity represents the effective intensity of the sampling spot corresponding to the sampling point, used to indicate the echo intensity of that point. It is expressed as Intensity, with a length of one byte (8 bits).

Intensity = (Si_2nd & 0x03) * 64 + (Si_L >> 2);

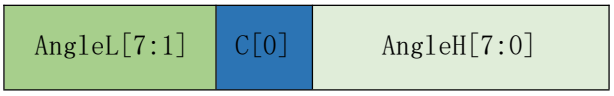
Distance Parsing:

Distance represents the ranging information corresponding to the sampling point, with a length of 14 bits and a unit of mm.

Distance = Si_H * 64 + (Si_2nd >> 2);

Angle Parsing:

Angle data is stored in FSA and LSA, representing the start angle and end angle respectively. Each angle data has the following data structure and uses little-endian mode.



Where C is the check bit with a fixed value of 1. The specific angle parsing process is as follows:

Start angle FSA calculation formula: Angle_fsa = (FSA >> 1) / 64;

End angle LSA calculation formula: Angle_lsa = (LSA >> 1) / 64;

Intermediate angle calculation formula: Angle(i) =

$$(FSA>>1)/64+\frac{(LSA>>1)/64-(FSA>>1)/64}{LSN-1}*(i-1);$$

[i=2,3...LSN-1]

Definition of high-reflectivity material flag bit:

The high-reflectivity flag indicates whether the current sampling point is made of a high-reflectivity material, represented as High Reflection. Bit[0] is the high-reflectivity material flag position, and Bit[1] is a reserved bit.

[0X01] indicates that the current sampling point is a high-reflectivity material, such as 3M reflective tape, metal, or other materials with a reflectivity exceeding 300%. [0X00] indicates materials with a reflectivity below 100%.

Checksum parsing:

The checksum uses a dual-byte XOR to verify the current data packet. It does not participate in the XOR operation itself, and the XOR order does not strictly follow the byte sequence; the XOR order is as shown in the diagram. It is important to note that since the sampling data Si consists of 3 bytes, the high 8 bits of the 3rd byte of Si must be padded with zeros before XORing, as shown in D5 and D6 in the figure below, where the MSB of D6 is padded with 0. Therefore, the checksum calculation formula is:

$$CS = \bigoplus_{i=1}^{n} D_i$$

This $\bigoplus_{i=1}^{n}$ This is the cumulative XOR formula, indicating that the elements D1, D4... Dn are XORed together. The XOR operation is independent of the order of elements.

LSB	MSB	
PHL	PHH	D1
M&T	LSN	D2
FSAL	FSAH	D3
LSAL	LSAH	D4
S1_L	S1_2nd	D5
S1_H	0x00	D6
S2_L	S2_2nd	D7
S2_H	0x00	D8
...		..
Si		Dn

revise

Date	Version	Revised content
2024-11-08	1.0	First Draft